NVIDIA

# Combining Walk-Text and Label Deconvolution Embeddings for SAGN-Based Product Node Classification

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## 01 WHY LD + WALK-TEXT?

**LD view:** task-oriented, graph-aware features from each product's attributes.
**Walk-text view:** descriptions of the root and visited products, read jointly by a frozen LM.
**Fusion:** keep both views plus raw features so SAGN/SLE can learn how to use them.

The **full pipeline** exceeds the shown published top score. [6, 7]

## LD: LABEL DECONVOLUTION [2]

LD learns graph-aware targets for text-encoder training.

**Two-node toy: headphones A - phone case D**

Scores [Audio, Accessories]; 75% own + 25% neighbor

| Case | A before | D before | A after mix |
|---|---|---|---|
| **Original** | [1, 0] | [0, 1] | [0.75, 0.25] |
| **Adjusted** | [1.5, -0.5] | [-0.5, 1.5] | [1, 0] |

**0.75 x 1.5 + 0.25 x (-0.5) = 1 (Audio)**
**0.75 x (-0.5) + 0.25 x 1.5 = 0 (Accessories)**
**Both predict Audio; adjusted scores cancel mixing.**

Toy scores, not probabilities or actual LD embeddings.
Real LD: nonnegative, normalized targets; labels stay fixed. [2]
Here: released **768-d LD-BERT embeddings**.

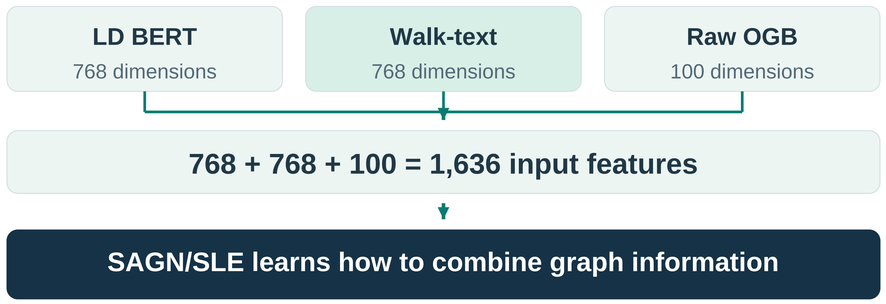
## 05 SAGN: THE GRAPH MODEL [3]

SAGN learns a hop mixture for each product.

| Product | 0-hop | 1-hop | 2-hop |
|---|---|---|---|
| A: Headphones | 80% | 15% | 5% |
| B: Earbuds | 70% | 25% | 5% |
| C: Cable | 40% | 40% | 20% |
| D: Phone case | 60% | 30% | 10% |
| **E: Sony WF-1000XM5** | **20%** | **60%** | **20%** |
| F: Charger | 65% | 25% | 10% |
| G: Power bank | 55% | 35% | 10% |
| H: Smartphone | 75% | 20% | 5% |

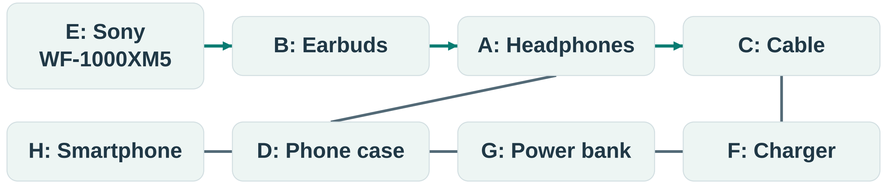
Toy hop weights; each row sums to 100%.

## 02 FUSE FEATURES + SAGN/SLE [3]




**Classifier:** SAGN uses the fused features (example at lower left).
**SLE:** four-stage training adds label propagation and confident pseudo-labels.

## 03 WALK-TEXT: ONE BIG EXAMPLE




**Root E: Sony WF-1000XM5 earbuds.** The walk **E - B - A - C** adds related-product context to its own description. Four nodes, three steps. Illustrative graph; these are not verified OGB items or co-purchase links.

**Schematic walk text: "Sony WF-1000XM5 earbuds" | "wireless earbuds" | "noise-cancelling headphones" | "USB cable"**

**Frozen EmbeddingGemma-300M**
160-token input | No encoder fine-tuning
Attention-mask mean pooling + L2 normalization

**768-d embedding for root E (Sony earbuds)**

**160 tokens:** four walks are generated, but only the first rendered walk's prefix is retained. No category labels enter the text. [6]

## 04 TEN-RUN RESULT [6]

# 87.4984%

**+/- 0.0490% test accuracy**

10 independent runs; mean +/- sample SD

**Validation: 94.0584% +/- 0.0433%**

## OUR RESULT + PUBLISHED OGB ROWS [7]

| Our model (unranked) | Test accuracy | Validation |
|---|---|---|
| **LD + frozen walk-text + SAGN/SLE + C&S** | **0.874984 +/- 0.000490** | **0.940584 +/- 0.000433** |

Unaltered OGB screenshot. Margin vs top entry: **+0.1284 pp**.

The classification accuracy on the test and validation sets. The higher, the better.

Package: >=1.1.1

| Rank | Method | Ext. data | Test Accuracy | Validation Accuracy | Contact | References | #Params | Hardware | Date |
|---|---|---|---|---|---|---|---|---|---|
| 1 | **GLEM+GIANT+SAGN+SCR** | Yes | 0.8737 ± 0.0006 | 0.9400 ± 0.0003 | Jianan Zhao (Mila & MSRA Team) | Paper, Code | 139,792,525 | Tesla V100 (32GB) | Oct 27, 2022 |
| 2 | **LD+GIANT+SAGN+SCR** | Yes | 0.8718 ± 0.0004 | 0.9399 ± 0.0002 | Zhihao Shi (MIRA Lab, USTC & CityBrain Lab, Alibaba Cloud) | Paper, Code | 110,636,896 | GeForce RTX 3090 (24GB GPU) | Sep 27, 2023 |
| 3 | **GraDBERT+GIANT & SAGN+SLE+CnS ** | Yes | 0.8692 ± 0.0007 | 0.9371 ± 0.0003 | Costas Mavromatis (UMN & AWS) | Paper, Code | 1,154,654 | GeForce RTX 3090 (24GB GPU) | Apr 20, 2023 |

## BENCHMARK & TASK [5]

**2.45M** products
**61.86M** undirected links
**47** categories

**ogbn-products**: Amazon co-purchase graph
Official sales-rank split
**196,615** training nodes
**39,323** validation; **2,213,091** test
External data: **Yes**

Task: predict one of 47 product categories using product attributes and graph relationships.

**WalkLM-style:** original WalkLM fine-tunes its LM; we use its walk-to-text idea with a frozen encoder. Our pipeline combines existing methods. [1, 6]

**Result qualification:** full pipeline; not an official rank/significance claim. V1 was test-aware; V2 locked validation before test. Artifact checks are not fresh retraining. [6]

[1] Tan et al. WalkLM, NeurIPS 2023. Original paper
[2] Shi et al. Label Deconvolution. arXiv:2309.14907
[3] Sun et al. SAGN + SLE. arXiv:2104.09376
[4] Huang et al. Correct & Smooth, ICLR 2021. arXiv:2010.13993
[5] OGB products dataset and official split
[6] Project protocol and final result evidence
[7] Official OGB leaderboard, checked 1 November 2026

Stanford Data-Centric AI Workshop, 3 November 2026 | mshanthappa@nvidia.com

**CODE + PROTOCOL**

Public repository
MIT code license

mshanthappa-hub/ld-walklm